
LECTURE NOTES

Lecture 9: Population Based Methods

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Contents

1	Population Methods	3
1.1	Genetic algorithms	3
1.1.1	Chromosomes	4
1.1.2	Selection	4
1.1.3	Crossover/Recombination	6
1.1.4	Mutation	6
1.2	Differential Evolution	6
1.3	Particle Swarm Optimization	7
1.4	Hybrid Methods	8

1 Population Methods

Instead of optimizing a single design point, population methods optimize a collection of **individuals**. We can increase the number of individuals to prevent the algorithm from being stuck in a local minimum

- Population methods begin with an initial population
- The initial population are usually uniform, normal distribution, or Cauchy distribution

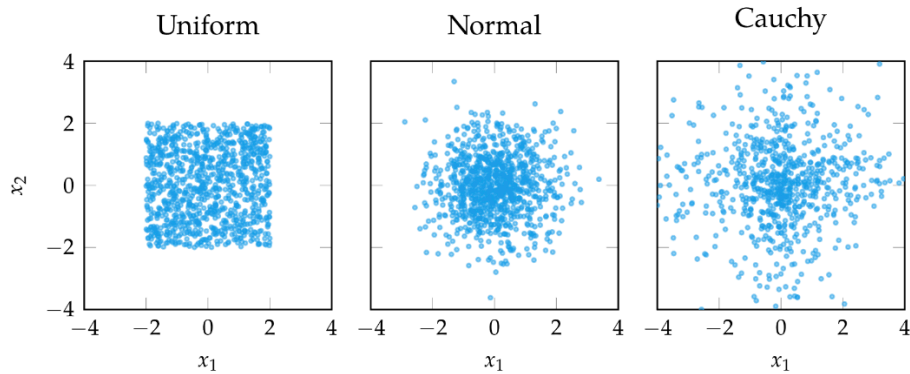


Figure 1: Different distributions for initializations of a population

1.1 Genetic algorithms

Inspired by biological evolution where the fittest individuals pass their genetic information to the next generation

- Here individuals are interpreted as chromosomes
- The fittest individuals are determined by **selection**
- The next generation is formed by selecting the fittest individuals and performing **crossover** and **mutation**

Algorithms like this are called **Darwinian Evolution methods**

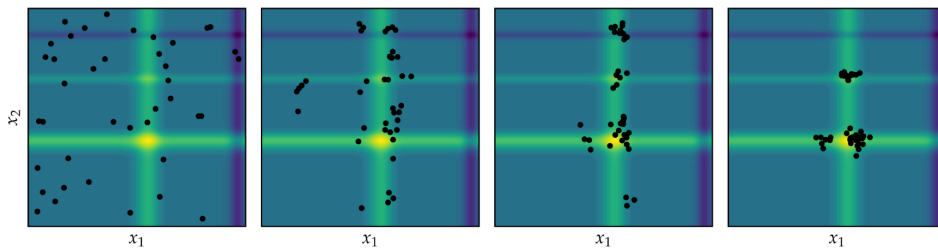


Figure 2: Genetic algorithm with truncation selection, single point crossover, and Gaussian mutation applied to Michalewicz function

$$f(x) = - \sum_{i=1}^d \sin(x_i) \sin^{2m} \left(\frac{ix_i^2}{\pi} \right)$$

1.1.1 Chromosomes

Chromosomes are typically initialized randomly (like a distribution)



Figure 3: Simple representation of chromosomes as a binary string. In this case a chromosome can either be 1 or 0 (red or blue)

Chromosomes are more commonly represented as real-valued chromosomes which are simply real-valued vectors

1.1.2 Selection

This processes determines which individuals pass their genetic information on to the next generation choosing chromosomes to use as parents for the next generation

- **Truncation selection:** kill the lowest performers (best fit score)
- **Tournament selection** selects fittest out of k randomly chosen individuals
- **Roulette Wheel selection** individuals are chosen with probability proportional to their fitness

$$p = \frac{\rho_i}{\sum_i \rho_i}$$

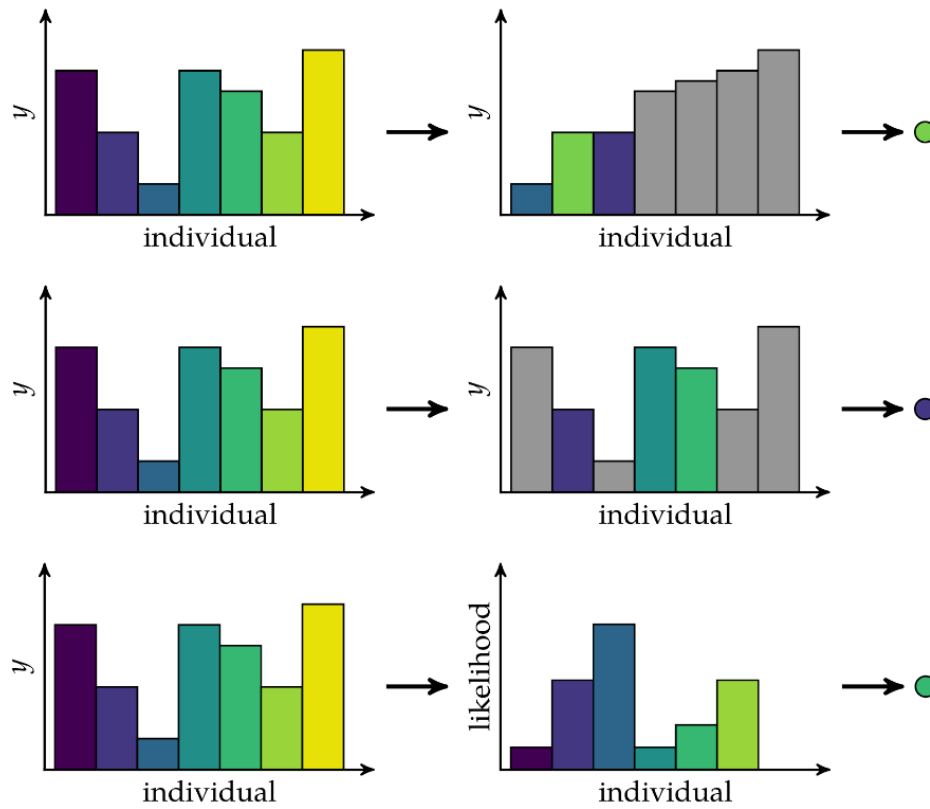
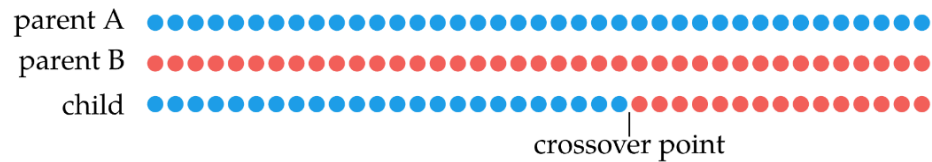


Figure 4: Exmaple of selection. Note that for this exmaple $\frac{1}{\text{fitness}} = 1 - \frac{1}{y}$

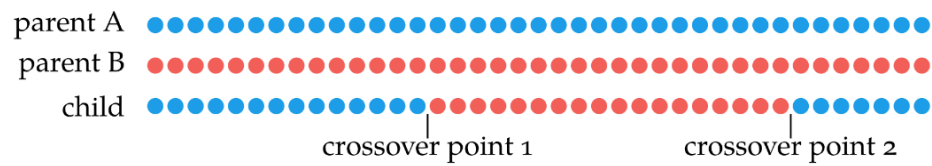
1.1.3 Crossover/Recombination

Combines the chromosomes of the parents to form children

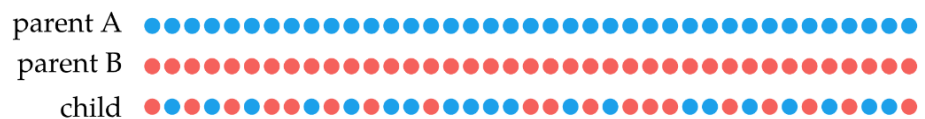
- Single-point crossover: swap occurs after single crossover point



- Two-point crossover: two crossover points



- Uniform crossover: each bit has 50% chance of crossover



1.1.4 Mutation

Mutation supports exploration of new areas of design space

Each bit or real-valued element has a probability (**mutation rate**) of being flipped or modified by noise

1.2 Differential Evolution

Improves each individual x by recombining other individuals according to a simple formula

- 1) Choose three random, distinct individuals a , b , and c
- 2) Construct interim design $z = a + w(b - c)$
- 3) Choose a random dimension to optimize in
- 4) Construct candidate x' via binary crossover of x' and z

$$\begin{cases} z_i & i = j \text{ or with probability } p \\ x_i & \text{otherwise} \end{cases}$$

- 5) Insert better design between x and x' into next generation

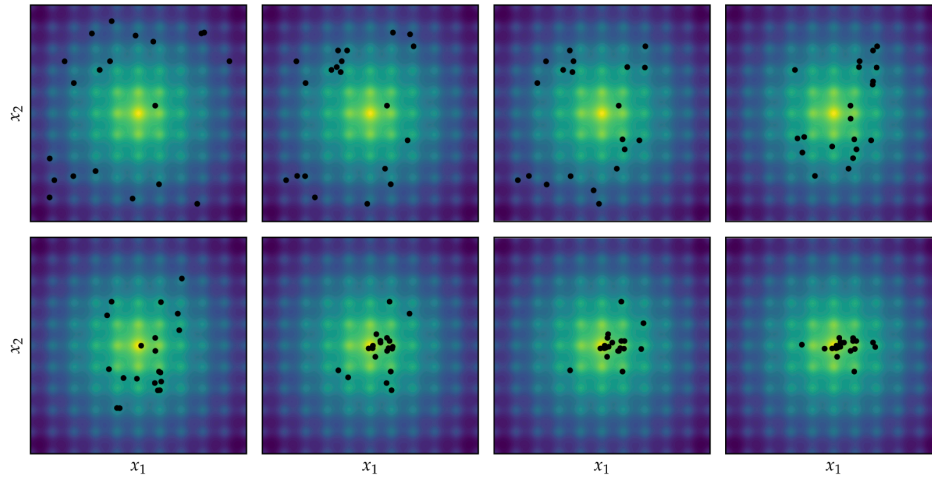


Figure 5: Example of differential evolution

1.3 Particle Swarm Optimization

Each individual, or particle, tracks the following

- Current position
- Current velocity
- Best position seen so far by the particle
- Best position seen so far by any particle

At each iteration, these factors produce **force** and **momentum** effects to determine the movement of a particle p from the population:

$$x_i^p \leftarrow x_i^p + v_i^p$$

$$v_i^p \leftarrow wv_i^p + c_1r_1(x_i^{p, \text{best}} - x_i^p) + c_2r_2(x_i^{\text{best}} - x_i^p)$$

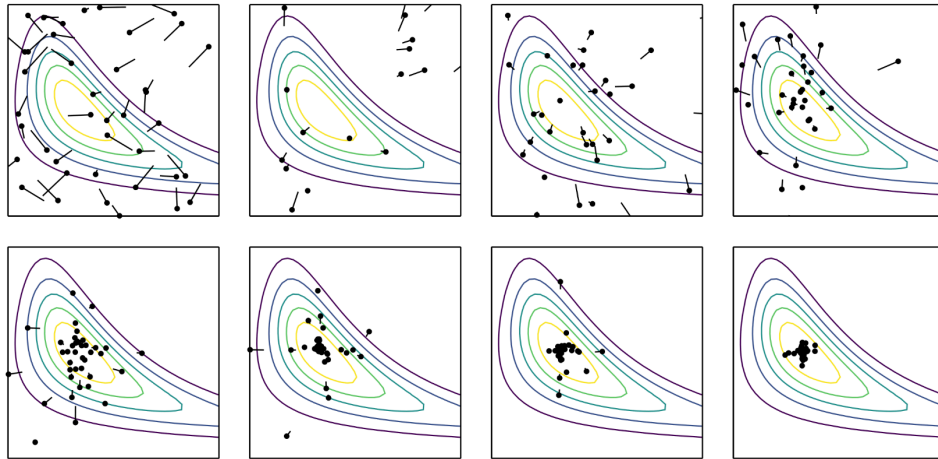


Figure 6: Example of particle swarm optimization

There are different kinds of swarm algorithms: <https://fcampelo.github.io/EC-Bestiarly/>

1.4 Hybrid Methods

Generally, these methods (Darwinian Evolution methods) are good at finding the best regions in design space, but do not perform as well as descent methods near the minimizer

(Some giraffes are born with longer necks. They access more food. They reproduce more. Long-neck genes spread.)

Hybrid methods try to leverage the strength of both methods

Here are two hybrids: